MILITARY SPECIFICATION

FOR

ACTUATOR, ELECTRO-PYROTECHNIC, LINEAR,

ATU-89/B

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the ATU-89/B Linear Electro-Pyrotechnic Actuator critical item, referred to herein as the actuator.

2. APPLICABLE DOCUMENTS.

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-I-23659	Initiator, Electric, Design and Evaluation of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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MIL-C-45662 — Calibration System Requirements.

STANDARDS

FEDERAL

FED-STD-102 — Preservation, Packaging, and Packing Levels.

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

MIL-STD-202 — Test Methods for Electronic and Electrical Component Parts.

MIL-STD-331 — Fuze and Fuze Components, Environmental and Performance Tests for.

MIL-STD-454 — Standard General Requirements for Electronic Equipment.

MIL-STD-810 — Environmental Test Methods.

MIL-STD-831 — Test Reports, Preparation of.

MIL-STD-1167 — Ammunition Data Cards.

PUBLICATIONS

REPORTS

Naval Sea Systems Command (Code Ident 10001)

NAVORD Report 2101 — Statistical Method Appropriate for Evaluation of Fuze Explosive Train Safety and Reliability.

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DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

DL 1155AS570	Actuator, Electro-Pyrotechnic, Linear, ATU-89/B, Data List.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-178	Transportation.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS.

3.1 Item definition. The actuator is an electrically initiated pyrotechnic device which is rated with a no-fire capability of 1 ampere (A), 1 watt (W) (1 A/1 W) (see Figure 1).

3.2 Characteristics.

3.2.1 Performance.

3.2.1.1 Electrical.

3.2.1.1.1 Electrostatic discharge. The actuator shall not fire when subjected to an electrostatic discharge of 25,000 ±500 volts (V) applied between the bridgewire leads and between both bridgewire leads shorted together and the actuator case.

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3.2.1.1.2 Dielectric strength. The actuator shall not fire during application of 500 volts direct current (Vdc) between both bridgewire leads shorted together and the actuator case for a duration of 60 seconds (s) minimum. The leakage current shall not exceed 0.1 milliamperes (mA) maximum.

3.2.1.1.3 No-fire sensitivity. The actuator shall not fire when a direct current of 1.00 ±0.05 A is applied to the bridgewire circuit for 30 milliseconds (ms) minimum.

3.2.1.1.4 Bridgewire resistance. The bridgewire circuit shall have a resistance of 1.00 ±0.15 ohms.

3.2.1.2 Seal. The leak rate of the seal shall not exceed 1×10^{-6} standard atmospheric cubic centimeters per second.

3.2.1.3 Neutron radiographic. The actuator shall show no visual evidence of loose powder, voids, or fissures in the charge or insulators during neutron radiographic inspection.

3.2.1.4 Functional.

3.2.1.4.1 All-fire sensitivity.

3.2.1.4.1.1 All-fire, 5 A. The actuator shall fire when the bridgewire circuit is subjected to a 5.0 ±0.1 A input and shall fully extend the piston 0.25 inch minimum, with an initial force of 100 pounds (lb) minimum and a final force of 20 lb minimum.

3.2.1.4.1.2 All-fire, 16 Vdc. The actuator shall fire when the bridgewire circuit is subjected to a 16 ±2 Vdc input and shall fully extend the piston 0.25 inch minimum, with an initial force of 100 lb minimum and a final force of 20 lb minimum.

3.2.1.4.1.3 All-fire Bruceton. The actuator shall have a calculated 99 percent all-fire point of 5 A maximum at a confidence level of 95 percent.

3.2.1.4.1.4 Resistance after firing. After firing, the resistance between the bridgewire leads and between both bridgewire leads shorted together and the actuator case shall be 20 kilohms minimum when measured at 20 ±2 Vdc.

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3.2.1.4.2 No-fire sensitivity.

3.2.1.4.2.1 No-fire, 1 A/1 W, 5 minutes. The actuator shall not fire when subjected to 1 A or 1 W for a 5-minute minimum time span.

3.2.1.4.2.2 No-fire Bruceton. The actuator shall have a calculated 1.0 percent no-fire point of 1 A minimum at a confidence level of 95 percent.

3.2.2 Environmental conditions. The actuator shall not be damaged, nor shall safety or subsequent performance be degraded, by exposure to any natural combination of service-use environments. These shall include, but shall not be limited to, the following environments:

3.2.2.1 Forty-foot drop (destructive). The actuator shall meet the requirements of MIL-STD-331, Test 103.1, Procedure 1. The actuator shall not fire during the test and shall be safe to handle and dispose of after the test.

3.2.2.2 Shock (nondestructive). The actuator shall meet the shock test requirements of MIL-I-23659. The actuator shall not fire when subjected to a 200-gravity unit (g) shock. The bridgewire resistance shall be as specified in 3.2.1.1.4, and the actuator shall function as specified in 3.2.1.4.1.1 or 3.2.1.4.1.2 after the test.

3.2.2.3 Temperature-shock/humidity/altitude (nondestructive). The actuator shall meet the temperature-shock/humidity/altitude test requirements of MIL-I-23659 for a 14-day period. The bridgewire resistance shall be as specified in 3.2.1.1.4, and the actuator shall function as specified in 3.2.1.4.1.1 or 3.2.1.4.1.2 after the test.

3.2.2.4 Vibration (nondestructive). The actuator shall meet the requirements of MIL-STD-810, Method 514.2, Equipment Category e, Procedure V, Part 2, Table 514.2-V, Figure 514.2-5, Curve AE. The bridgewire resistance shall be as specified in 3.2.1.1.4, and the actuator shall function as specified in 3.2.1.4.1.1 or 3.2.1.4.1.2 after the test.

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3.3 Design and construction.

3.3.1 Production drawings. The actuator shall be fabricated and assembled in accordance with the drawings and documents listed on DL 1155AS570 and as specified herein.

3.3.2 Standards of manufacture. The actuator shall be manufactured, inspected, and tested to meet the requirements specified herein.

3.3.3 Workmanship. Workmanship during manufacture of the actuator shall meet the requirements of MIL-STD-454, Requirement 9. The requirement for first class workmanship shall not be limited solely to the preceding standard and shall not preclude the use of time- and labor-saving techniques, provided that safety and reliability will in no way be jeopardized.

3.4 Test plan and test procedures. When specified in the contract or purchase order and using MIL-STD-831 as a guide, the contractor shall prepare and submit to the Government for approval a test plan and test procedures in accordance with the applicable Data Item Description cited in 6.2.2.

3.5 Standard test equipment calibration. All test equipment shall be calibrated and maintained in accordance with MIL-C-45662. When specified in the contract or purchase order, the contractor shall prepare and submit to the Government for approval a calibration plan and operation maintenance procedures in accordance with the applicable Data Item Description cited in 6.2.2.

3.6 First article.

3.6.1 Preproduction sample. Unless otherwise specified in the contract or purchase order, the contractor shall deliver a preproduction sample of 250 actuators to the testing activity designated in the contract or purchase order for preproduction testing (see 6.2.1). The preproduction sample shall be manufactured using the same methods, materials, processes, and procedures proposed for production. Any production prior to acceptance of the preproduction sample is at the risk of the contractor.

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4. QUALITY ASSURANCE PROVISIONS.

4.1 General.

4.1.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements (see 6.2.1).

4.1.1.1 Classification of inspections. Inspection requirements specified herein are classified as follows:

(a) Preproduction inspections (see 4.1.2.1).

(b) Quality conformance inspections (see 4.2).

(c) In-process inspections (see 4.1.2.1, 4.2 and 4.4.1).

4.1.2 Special tests and examinations.

4.1.2.1 Preproduction inspections. Prior to the environmental tests, the preproduction sample (see 3.6.1) shall be subjected to the in-process inspections of Table I, Group I. The preproduction sample shall then be subjected to the inspections of Table I, Groups II through X in the sequence shown. Failure of the preproduction sample to meet any of the requirements specified herein shall cause rejection of the preproduction sample.

4.2 Quality conformance inspections. Prior to formation of the production lot, each actuator shall be subjected to the in-process inspections of Table II, Group A. The inspection lot (see 4.2.1) shall then be subjected to the tests of Table II, Groups B through F. Failure of any sample actuator to meet the requirements specified herein shall cause rejection of the lot represented.

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TABLE I. Preproduction sample inspection sequence.

Inspection	Requirement paragraph	Method paragraph
Group I (250 actuators) (in-process)		
Drawing requirements	3.3.1, 3.3.3	4.4.1
Electrical	3.2.1.1	4.4.2
Seal	3.2.1.2	4.4.3
Neutron radiographic	3.2.1.3	4.4.4
Group II (6 actuators)		
Shock	3.2.2.2	4.4.6.2
Bridgewire resistance	3.2.1.1.4	4.4.2.4
All-fire, 16 Vdc (information only)	3.2.1.4.1.2	4.4.5.1.2
Group III (6 actuators)		
Forty-foot drop	3.2.2.1	4.4.6.1
Group IV (60 actuators)		
Temperature-shock/humidity/altitude	3.2.2.3	4.4.6.3
Group V (20 actuators)		
Vibration	3.2.2.4	4.4.6.4
Group VI (60 actuators from Group IV, 20 from Group V, plus 58 spares)		
Bridgewire resistance	3.2.1.1.4	4.4.2.4
Dielectric strength	3.2.1.1.2	4.4.2.2
Group VII (69 actuators from Group VI)		
No-fire, 1 A/1 W	3.2.1.4.2.1	4.4.5.2.1
All-fire, 16 Vdc	3.2.1.4.1.2	4.4.5.1.2
Resistance after firing	3.2.1.4.1.4	4.4.5.1.4
Group VIII (69 actuators from Group VI)		
No-fire, 1 A/1W	3.2.1.4.2.1	4.4.5.2.1
All-fire, 5 A	3.2.1.4.1.1	4.4.5.1.1
Resistance after firing	3.2.1.4.1.4	4.4.5.1.4

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TABLE I. Preproduction sample test sequence. (Continued)

Inspection	Requirement paragraph	Method paragraph
Group IX (50 actuators)		
No-fire Bruceton	3.2.1.4.2.2	4.4.5.2.2
Group X (50 actuators)		
All-fire Bruceton	3.2.1.4.1.3	4.4.5.1.3
Resistance after firing	3.2.1.4.1.4	4.4.5.1.4

TABLE II. Production lot inspection sequence.

Inspection	Requirement paragraph	Method paragraph
Group A (in-process)		
Drawing requirements	3.3.1, 3.3.3	4.4.1
Electrical	3.2.1.1	4.4.2
Seal	3.2.1.2	4.4.3
Neutron radiographic	3.2.1.3	4.4.4
Group B		
Forty-foot drop	3.2.2.1	4.4.6.1
Group C		
Vibration	3.2.2.4	4.4.6.4
Bridgewire resistance	3.2.1.1.4	4.4.2.4
Dielectric strength	3.2.1.1.2	4.4.2.2
Group D (actuators from Group C)		
No-fire, 1 A/1 W	3.2.1.4.2.1	4.4.5.2.1
All-fire, 5 A	3.2.1.4.1.1	4.4.5.1.1
Resistance after firing	3.2.1.4.1.4	4.4.5.1.4
Group E (actuators from Group C)		
No-fire, 1 A/1 W	3.2.1.4.2.1	4.4.5.2.1
All-fire, 16 Vdc	3.2.1.4.1.2	4.4.5.1.2
Resistance after firing	3.2.1.4.1.4	4.4.5.1.4

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TABLE II. Production lot inspection sequence. (Continued)

Inspection	Requirement paragraph	Method paragraph
Group F		
Verify shorting of lead wires	Section 5	4.5
Packaging, packing, and marking	Section 5	4.5

Lot size (actuators)	Sample size (actuators)	Quantity of actuators for groups					
		A	B	C	D	E	F
151 to 500	32	All	6	26	10	16	Remainder of lot
501 to 3,200	125	All	24	101	40	61	
3,201 to 10,000	200	All	42	168	60	108	

4.2.1 Inspection lot. Unless otherwise specified in the contract or purchase order, an inspection lot (see 6.2.1) shall contain not less than 151 nor more than 10,000 actuators, including those required for destructive tests. An inspection lot, as defined in MIL-STD-105, shall consist of actuators fabricated from the same materials, using the same processes and procedures under the same conditions, and offered for inspection at the same time.

4.2.2 Sampling. Unless otherwise specified in the contract or purchase order (see 6.2.1), the sample lot size specified in Table III, based on the inspection lot size, shall be randomly selected for sampling. The inspection lot shall be rejected if the number of defects exceeds the number allowed for a specified inspection lot size.

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TABLE III. Sampling.

Inspection lot size	Sample lot size	Number of defects	
		Accept on	Reject on
151 to 500	32	0	1
501 to 3,200	125	1	2
3,201 to 10,000	200	2	3

4.2.2.1 Defects. A defect exists when one or more of the following conditions occurs in a single actuator:

(a) When any actuator functions other than when specified in the functional tests of 4.4.5.1.

(b) When any actuator fails to fire on the first firing attempt, except when firing for information only.

(c) When one or more of the requirements of Section 3 are not met.

4.3 Test equipment and conditions.

4.3.1 Standard test equipment. The contractor shall provide and maintain an adequate system of inspection and test equipment necessary to ensure that parts and products will meet the contract or purchase order, specification, and drawing requirements. Test equipment, including associated calibration plans and operation and maintenance procedures, shall be approved by the procuring activity (see 3.5). When special test equipment and circuits are devised or commercially available equipment is employed, all test equipment circuits and methods are subject to approval by the procuring activity. Unless otherwise specified herein, the magnitude of any error introduced by test equipment shall not exceed 10 percent of the tolerance of the requirement being measured.

4.3.2 Test conditions. Unless otherwise specified herein, the test methods of 4.4 shall be performed under the following conditions:

(a) Temperature: Room ambient of 65 ±5 degrees Fahrenheit (°F) to 95 ±5°F.

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(b) Altitude: Normal ground elevation.

(c) Humidity: Room ambient to 40 percent relative humidity maximum.

4.4 Methods of inspection.

4.4.1 Drawing inspections. The drawing inspections shall be performed on each lot of parts or subassemblies in accordance with DL 1155AS570.

4.4.2 Electrical. To determine conformance with 3.2.1.1, the following electrical tests shall be performed:

WARNING

The actuator is an explosive-loaded device; suitable safety precautions shall be observed (see 6.3).

4.4.2.1 Electrostatic discharge. To determine the sensitivity to electrostatic discharge, connect a 500 ±5 picofarad (pF) capacitor charged to 25,000 ±500 V, and a 5,000 ohm ±5 percent resistor in a series circuit between the two shorted bridgewire leads and the actuator case. The series connection shall be maintained for 60 s minimum. Switching in this circuit shall be accomplished by closing a vacuum relay. The actuator used for this test shall be temperature-conditioned at 70 ±5°F for a period of 12 hours minimum. To be acceptable, the actuator shall meet the requirements of 3.2.1.1.1.

4.4.2.2 Dielectric strength. To determine dielectric strength, apply a voltage of 500 ±25 Vdc for 60 s minimum between both bridgewire leads shorted together and the actuator case. To be acceptable, the actuator shall meet the requirements of 3.2.1.1.2.

4.4.2.3 No-fire sensitivity. To determine the no-fire sensitivity, apply 1.00 ±0.05 A direct current through the bridgewire circuit for 30 ms minimum. To be acceptable, the actuator shall meet the requirements of 3.2.1.1.3.

4.4.2.4 Bridgewire resistance. To determine bridgewire resistance, mount the actuator in a physically-shielded fixture

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and measure the resistance between the two bridgewire leads with an instrument having an accuracy of 0.05 ohm in the range of 0.1 to 2.0 ohms and current limited to 10 mA. To be acceptable, the actuator shall meet the requirements of 3.2.1.1.4.

4.4.3 Seal. The actuator shall be tested in accordance with MIL-STD-202, Method 112B, Procedure IIIa, Test Condition C. To be acceptable, the actuator shall meet the requirements of 3.2.1.2.

4.4.4 Neutron radiographic. Using neutron radiographic equipment and procedures that will provide detection of loose powder, voids, fissures in compressed charges, and fissures or voids in insulating material, perform neutron radiographic inspections on each actuator, in groups if desired, along an axis perpendicular to the longitudinal axis. The radiographic plates shall identify the part number, lot number, and test date. Unless otherwise specified in the contract or purchase order, equipment and procedures shall be approved by the procuring activity prior to use (see 6.2.1). To be acceptable, the actuator shall meet the requirements of 3.2.1.3.

4.4.5 Performance tests. All actuators of the preproduction sample, or a randomly selected sample from the production lot, shall be subjected to the tests of Table I or Table II, as applicable. All actuators subjected to the tests of Groups II through X of Table I, or Groups B through F of Table II shall be fired during the test sequence and not returned to the lot nor forwarded for service use.

4.4.5.1 Functional. The actuator shall be installed in a suitable test fixture to determine the output force and piston extension specified in 3.2.1.4. The functional tests shall be conducted as follows:

4.4.5.1.1 All-fire, 5 A. Apply 5.0 ±0.1 A through the bridgewire leads. To be acceptable, the actuator shall meet the requirements of 3.2.1.4.1.1 and, subsequently, the requirements of 3.2.1.4.1.4.

4.4.5.1.2 All-fire, 16 Vdc. Apply 16 ±2 Vdc across the bridgewire leads. To be acceptable, the actuator shall meet the requirements of 3.2.1.4.1.2 and, subsequently, the requirements of 3.2.1.4.1.4.

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4.4.5.1.3 All-fire Bruceton. Subject 50 actuators from the preproduction sample to the all-fire Bruceton methods of NAVORD Report 2101 to determine the least firing stimulus which causes initiation within 50 ms maximum. The current shall be applied to the bridgewire circuit. Actuators used for this test shall be pulsed with a varying current level for 50 ms maximum. The initiation response shall be determined by proper extension of the piston. To be acceptable, the actuator shall meet the requirements of 3.2.1.4.1.3.

4.4.5.1.4 Resistance after firing. After firing the actuator, the resistance between the bridgewire leads, and the resistance between both bridgewire leads shorted together and the actuator case shall be measured. This may be accomplished by applying a 20 ±2 Vdc potential. To be acceptable, the actuator shall meet the requirements of 3.2.1.4.1.4.

4.4.5.2 No-fire sensitivity.

4.4.5.2.1 No-fire, 1 A/1 W, 5 minutes. Apply a direct current of 1 A minimum supplying 1 W minimum to the bridgewire circuit for a period of 5 minutes minimum. The test current shall be held constant throughout the 5-minute period to within 2 percent. The actuator shall be preconditioned at 70 ±5°F for 12 hours. To be acceptable, the actuator shall meet the requirements of 3.2.1.4.2.1.

4.4.5.2.2 No-fire Bruceton. Subject 50 actuators from the preproduction sample to the no-fire Bruceton method of NAVORD Report 2101 to determine the greatest firing stimulus which does not cause initiation within 5 minutes maximum. Fifteen actuators shall be used in search of the mean, and the remaining 35 shall be used to determine the maximum no-fire current. This test shall be conducted at room ambient temperature with no heat sink. To be acceptable, the actuator shall meet the requirements of 3.2.1.4.2.2.

4.4.6 Environmental.

4.4.6.1 Forty-foot drop (destructive). The actuator shall be tested in accordance with MIL-STD-331, Test 103.1, Procedure 1. To be acceptable, the actuator shall meet the requirements of 3.2.2.1.

4.4.6.2 Shock. The actuator shall be mounted in a test fixture which simulates service-use conditions, and subjected to a 200 g ±10 percent shock pulse along each of the three mutually

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perpendicular axes in accordance with MIL-I-23659. The shape of each shock pulse shall approximate, as nearly as possible, a sine wave with 200 g ±10 percent for 1.5 ±0.4 ms and shall exceed 65 g for 9.0 ±0.9 ms. To be acceptable, the actuator shall meet the requirements of 3.2.2.2.

4.4.6.3 Temperature-shock/humidity/altitude (nondestructive). The actuator shall be mounted in a test fixture which simulates service-use conditions, and tested in accordance with MIL-I-23659, except the test duration shall be for a 14-day period. Following the test cycle, measure the bridgewire resistance in accordance with 4.4.2.4. To be acceptable, the actuator shall meet the requirements of 3.2.2.3.

4.4.6.4 Vibration (nondestructive). The actuator shall be mounted to a test fixture which simulates service-use conditions, and tested in accordance with MIL-STD-810, Method 514.2, Equipment Category e, Procedure V, Part 2, Table 514.2-V, Figure 514.2-5, Curve AE, except vibration shall be applied for a period of 2 hours minimum along each of the three mutually perpendicular axes for a total test time of 6 hours. To be acceptable, the actuator shall meet the requirements of 3.2.2.4.

4.5 Packaging, packing, and marking. Prior to shipment, examination shall be made to ensure that packaging, packing, and marking conform to the requirements of Section 5. The bridgewire leads shall be shorted together prior to shipment.

5. PACKAGING.

5.1 Preservation-packaging. Preservation and packaging shall be in accordance with FED-STD-102.

5.1.1 Level C. Actuators shall be clean, dry, and packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing.

5.2.1 Level C. Actuators, packaged in accordance with 5.1.1 and with lead wires shorted together, shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. Packing shall conform to 49 CFR 171-178. An ammunition data card in accordance with MIL-STD-1167 shall accompany each lot and serially-numbered actuator.

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5.3 Marking. In addition to any special marking required by the contract or purchase order, all marking shall be in accordance with MIL-STD-129 (see 6.2.1).

6. NOTES AND CONCLUDING MATERIAL.

6.1 Intended use. The actuator is intended for use as a component of the Fuel Air Explosive (FAE) II FSU-9/B and FSU-13/B Bomb Safety-Arming Device.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

(a) Number, title, and date of this specification.

(b) Preproduction testing activity (see 3.4.1).

(c) When a first article is required, it should be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulations (DAR). The first ariticle should be a preproduction sample consisting of 250 actuators as specified in 3.6.1. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the first article.

(d) Responsibility for inspection (see 4.1.1).

(e) Inspection lot (see 4.2.1).

(f) Sample lot size, if other than as specified in 4.2.2.

(g) Neutron radiographic equipment and procedures (see 4.4.4).

(h) Marking (see 5.3).

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6.2.2 Data requirements. When this specification is used in a procurement which incorporates a Contract Data Requirements List (DD Form 1423) and invokes the provisions of 7-104.9(n) of the DAR, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved DD Form 1423 incorporated in the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited as follows:

Paragraph	Data Requirement	Applicable DID
3.4	Test Plan and Test Procedures	DI-T-5204
3.5	Calibration Plan	UDI-T-23742A

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

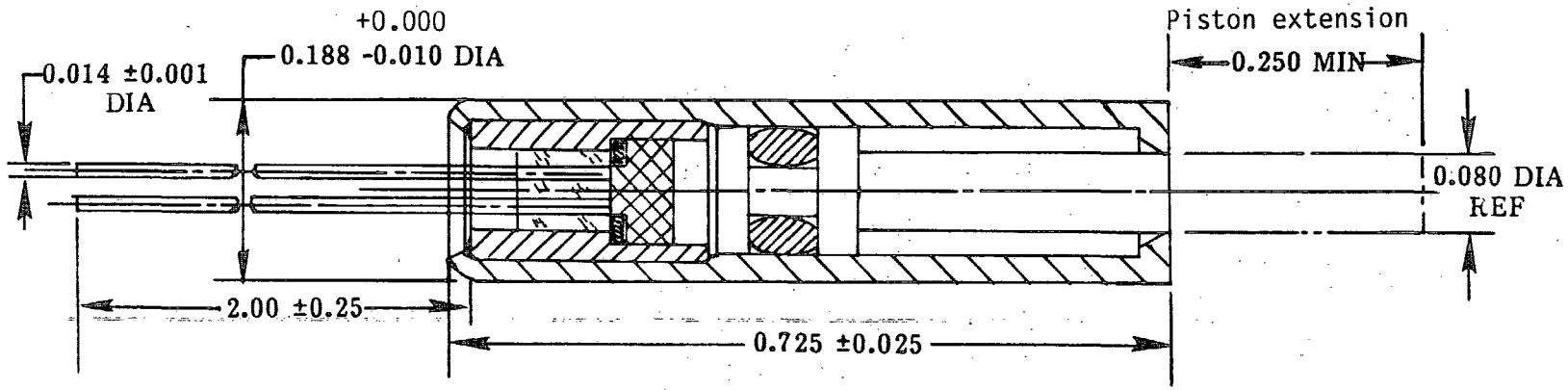
6.3 Explosives safety precautions. Minimum explosive safety precautions for use by the contractor are detailed in DoD Instruction 4145.26M, DoD Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material.

Preparing activity:

Navy - AS

(Project 1325-N100)

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NOTE. Dimensions are in inches.

FIGURE 1. ATU-89/B Linear Electro-Pyrotechnic Actuator.

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